HW: Labs 2, 22.10.26

2.5) Let $\Omega = [0,1]$ and let $\mathcal{F}_3$ consist of all countable subsets of Ω and all subsets of Ω having a countable complement.

a) Show that $\mathcal{F}_3$ is a σ-algebra.

$\mathcal{F}_3 = \{A \subset \Omega \mid A \text{ is countable or } A^c \text{ is countable}\}$

1) $\emptyset$ is countable $\Rightarrow \emptyset \in \mathcal{F}_3$

2) $A \in \mathcal{F}_3$, therefore either A is countable or A^c is countable

If A^c is countable, then by definition of $\mathcal{F}_3$, $A^c \in \mathcal{F}_3$

If A is countable, then $(A^c)^c = A$ is countable, therefore $A^c \in \mathcal{F}_3$

3) $\{A_i\}_{i=1}^{\infty}$, $A_i \in \mathcal{F}_3$, therefore A_i countable or A_i^c countable.

$\bigcup_{i=1}^{\infty} A_i =$

- All A_i countable $\Rightarrow \bigcup_{i=1}^{\infty} A_i$ countable $\Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}_3$
- At least one A_j uncountable $\Rightarrow \exists j$. A_j uncountable, but since $A_j \in \mathcal{F}_3$, A_j^c is countable. Thus $\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c \subset A_j^c$ countable. Since $\left(\bigcup_{i=1}^{\infty} A_i\right)^c \in \mathcal{F}_3$

so is $\bigcup_{i=1}^{\infty} A_i$. ▨

b) Let us define $P(A) = 0$ if A is countable, and $P(A) = 1$ if A has a countable complement. Is $(\Omega, \mathcal{F}_3, P)$ a legitimate probability space?

Well defined since A and A^C can't both be countable. The $A \cup A^C = [0,1]$ uncountable $\Rightarrow$ countable.

$P([0,1]) = 1$ since $[0,1]^C = \emptyset$ which is countable.

1) $P(\emptyset) = 0$ since $\emptyset$ is countable

2) $\{A_i\}_{i=1}^{\infty}$, $A_i \in \mathcal{F}_3$ (A_i countable or A_i^C countable), $\forall i \neq j : A_i \cap A_j = \emptyset$

$P(\bigcup_{i=1}^{\infty} A_i) =$

→ If all A_i countable, then $\bigcup_{i=1}^{\infty} A_i$ is countable, thus $0 = P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \underbrace{P(A_i)}_{0} = 0$ ✓

→ If at least one A_i is uncountable, then the union is uncountable. $1 = P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i) \geq 1$?

Assume $\exists j : A_j$ uncountable but $A_j \in \mathcal{F}_3$. Then A_j^C must be countable! But since $\forall i \neq j : A_i \cap A_j = \emptyset$, then $\forall i : A_i \in A_j^C$ which means that A_i must be countable! ← This means that $\sum_{i=1}^{\infty} P(A_i) = 1$!

$(\Omega, \mathcal{F}_3, P)$ is a legitimate prob. space! ∎